

Capstone Project Proposal

Project Title

NDI-Sentinel: The Autonomous Governance & Sync Agent

AI-Powered Autonomous Data Governance and NDI Compliance System

1. Problem Statement

Government entities and organizations struggle to continuously measure and maintain compliance with NDMO regulations and the National Data Index (NDI). Currently, compliance assessments are mostly manual, periodic, and dependent on spreadsheets, static reports, and human review.

This creates several problems:

- Governance policies exist in documents, while actual data is scattered across databases and files
- Sensitive data may remain undiscovered for long periods
- Evidence collection for SDAIA evaluations is slow and labor-intensive
- Organizations lack real-time visibility into their governance maturity level
- Manual audits increase human error and operational cost

The problem mainly affects:

- Data Management Offices (DMOs)
- Compliance teams
- IT and security departments
- Government organizations required to follow NDMO standards

The project aims to automate governance monitoring and provide continuous visibility into organizational compliance and NDI maturity.

2. Review of Current Solutions (Status Quo)

Currently, most organizations handle data governance and NDI compliance using manual and fragmented processes. Data Management Offices (DMOs) usually depend on:

- Excel sheets for tracking compliance
- Periodic manual audits
- Static PDF policies and reports
- Human review for evidence collection
- Traditional security and DLP tools

These approaches create a major gap between governance policies and the actual state of organizational data.

For example:

- Sensitive data may exist in databases or shared folders without being detected.
- Compliance checks are performed only every few months instead of continuously.
- Governance teams spend significant time preparing evidence for SDAIA assessments.
- Existing tools can detect issues, but they usually do not connect findings directly to NDI maturity indicators.

Current Evaluation Metrics

Organizations commonly evaluate governance performance using:

- Compliance accuracy
- Number of detected violations
- Audit preparation time
- Data classification accuracy
- False positive and false negative rates

For AI-based detection systems:

- Precision
- Recall
- F1-score
- mAP for object detection models

Limitations of Existing Solutions

Current solutions still face several limitations:

- Heavy dependence on manual work
- No real-time governance monitoring
- Limited automation in evidence generation
- Weak integration between policies and actual data assets
- Lack of intelligent remediation suggestions
- Most governance tools are expensive and complex to customize

3. Our Approach

High-Level Overview

NDI-Sentinel is a Multi-Agent AI Governance System designed to automate data governance assessment and compliance monitoring.

The system analyzes organizational data assets, compares them against NDMO governance policies, detects compliance gaps, and generates evidence reports automatically.

Inputs

The system receives:

- Database tables
- Organizational documents
- PDF policies
- Images and scanned files
- Metadata records
- Governance regulations and NDMO standards

Outputs

The system produces:

- Real-time NDI maturity scores
- Compliance violation reports
- Sensitive data detection alerts
- Auto-generated governance evidence
- Recommended remediation actions and policy-based guidance
- Governance dashboards and analytics
- Governance dashboards and analytics (Real-time NDI Maturity Tracking).

Agentic Workflow

The system uses a LangGraph-based multi-agent architecture consisting of:

1. Vision-Language Classification Agent

Responsible for:

- Detecting sensitive information in images and files
- OCR extraction
- Data classification
- Identifying unencrypted sensitive documents

Technologies:

- YOLOv8
- OCR
- NLP classifiers

2. RAG Policy Engine

Responsible for:

- Retrieving relevant NDMO policies
- Comparing organizational practices against governance requirements
- Performing gap analysis
- Producing grounded governance explanations

Technologies:

- LangChain
- LangGraph
- ChromaDB
- Local LLMs

3. NDI Scoring Agent & Dashboard Agent

- **Responsible for:** Translating compliance findings into governance maturity scores, tracking governance improvements over time, and feeding the real-time visual dashboard with NDI metrics and analytics.

4. Strategic Advisory Agent

Responsible for:

- **Gap Mapping:** Analyzing non-compliance issues and linking them to specific SDAIA/NDMO regulations.
- **Expert Recommendations:** Providing technical advice (e.g., encryption methods, anonymization) to bridge detected gaps.
- **Human-in-the-loop Support:** Generating actionable guidance for data stewards to ensure secure and compliant remediation.

5. Evidence Auto-Vault

Responsible for:

- Automatically generating PDF evidence files
- Creating compliance summaries
- Producing charts and statistics for SDAIA submissions

4. Data

Data Sources

1. NDMO Governance Documents

- Source: National Data Management Office (NDMO)
- URL:
- Author: NDMO / SDAIA

License: Public governmental documentation

2. Simulated Organizational Data

Generated datasets including:

- Employee records
- Customer databases
- Metadata tables
- Simulated sensitive documents
- Identity card images
- Unstructured files

License: Synthetic / internally generated data

Languages

- Arabic

Data Quality

- Official governance documents
- Recent and reliable sources
- Synthetic data to preserve privacy

5. Technologies

Category	Technologies
Orchestration	LangGraph
LLMs	Llama 3
Computer Vision	YOLOv8
OCR	EasyOCR
Vector DB	ChromaDB
Database	PostgreSQL
Backend	FastAPI
Deployment	Docker
Monitoring	LangFuse